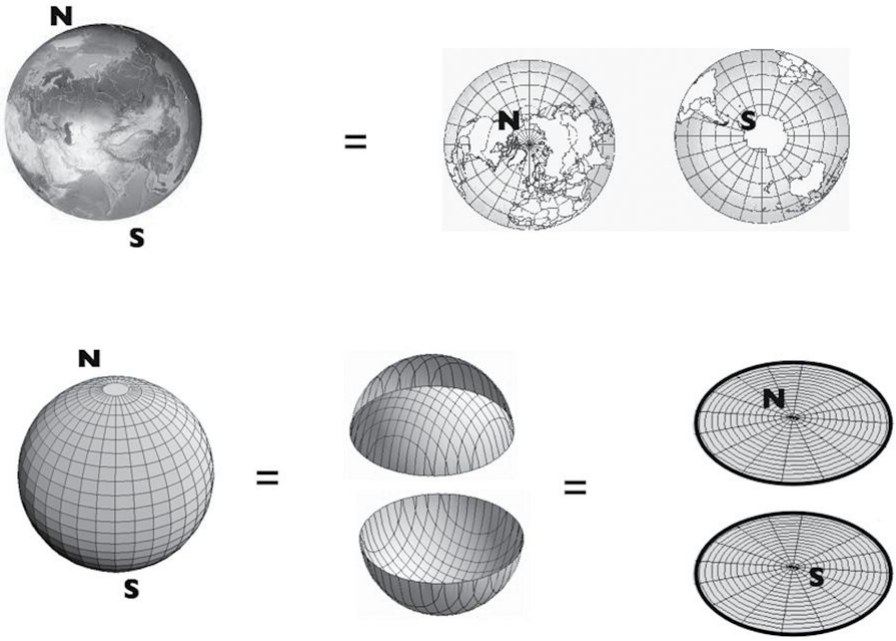


Figure 3.11 A sphere can be represented as two discs which in reality are smoothly joined all along their edges.

To understand the geometry of a ‘3-sphere’, let us return to the ordinary sphere; the surface of a ball, or the Earth. To represent the surface of the Earth on a plane, we can draw two discs, as is customary when drawing the continents (figure 3.11).

Notice that an inhabitant of the southern hemisphere is in a certain sense ‘surrounded’ by the northern hemisphere, since in whichever direction she goes to exit her hemisphere, she will always arrive in the other one. But the contrary is obviously true as well: each hemisphere ‘surrounds’, and is surrounded by, the other. A 3-sphere may be represented in a similar fashion, but with everything given an additional dimension: two balls stuck together all along their edges (figure 3.12).

When we leave one ball we enter into the other, just as when we leave one of the discs in the representation of the globe we enter into the other disc. Each ball surrounds and is surrounded by the other. Einstein’s idea is that space could be a 3-sphere: something with a finite volume (the sum of the volume of the two balls), but without borders.[fn17] The 3-sphere is the solution which Einstein proposes in his work of 1917 to the problem of the border of the universe. This article initiates modern cosmology, the study of the entire visible universe, studied at the grandest scale. From it will arise the discovery of the expansion of the universe; the theory of the Big Bang; the problem of the birth of the universe, and much else besides. I speak about all of this in Chapter 8.